Exponential Functions and Models

Concept

Exponential models multiply by a constant factor over equal time intervals.

Parent graph

$y=b^x$, $b>0$, $b\neq1$: domain all real x, range $y>0$, asymptote $y=0$.

Growth / decay

Multipliers $1+r$ / $1-r$; match the rate's time period.

Solve

Rewrite with the same base, or take logs after isolating the exponential.

Half-life

$A(t) = A_0(1/2)^{t/H}$, with half-life H.

Example 1

Solve $4^{x-1} = 8$.

Solution

1. Write powers of 2: $2^{2x-2} = 2^3$.

2. Equate exponents: $2x - 2 = 3 \rightarrow x = 5/2$.

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Example 2

A 160 g sample has a half-life of 3 hours. Find its mass after 9 hours.

Solution

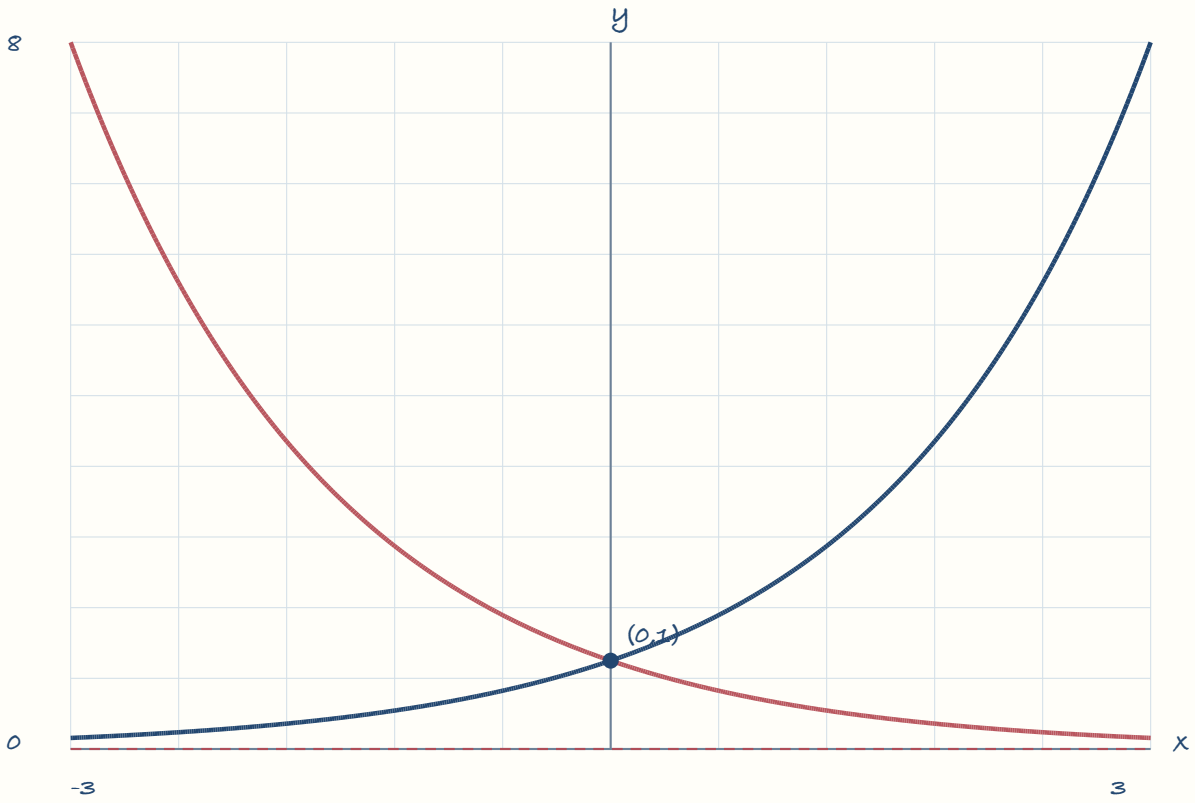
1. Number of half-lives $= 9/3 = 3$.

2. Mass $= 160(1/2)^3 = 20$ g.

Exponential Functions and Models

Visual notes | read the graph

Blue: $y = 2^x$; red: $y = (1/2)^x$



Both stay positive. One grows and the other decays. The horizontal asymptote is $y = 0$.

Try the picture

Locate intercepts and describe where the graph increases or decreases. Use the printed equation to verify at least two points. Explain any restrictions before extending the graph.

Exponential Functions and Models

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Solve $9^x = 27$.

2. Write a model for \$500 losing 12% of value each year.

3. Find the mass after 8 h if 96 g has a half-life of 4 h.

4. State domain, range and asymptote of $y = 5 \times 2^x - 3$.

Exponential Functions and Models

Practice answers | check after trying

1. $3^{2x} = 3^3$, so $x = 3/2$.

2. $V(t) = 500(0.88)^t$, $t \geq 0$ in years.

3. Two half-lives: $96/4 = 24$ g.

4. Domain all real x, range $y > -3$, asymptote $y = -3$.